# Sligo Creek Watershed Digital Twin

Model description, data and validation

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**Abstract**

The Sligo Creek digital twin (`https://openhydrotwin.com/SligoCreek/`) forecasts streamflow, stage, soil moisture and groundwater across the watershed for the next seven days, every six hours, with a 95 % band for the uncertainty of the calibrated parameters. This document describes the model behind it, a physically based, semi-distributed model of Sligo Creek, an urban tributary of the Northwest Branch of the Anacostia River in Montgomery and Prince George's Counties, Maryland, built in the open-source OpenHydroQual (OHQ) framework (Massoudieh et al., 2023). The model covers the whole 29.2 $\mathrm{km}^2$ watershed in 27 sub-catchments, each representing its impervious and pervious land, three soil layers, a regional aquifer and a near-stream aquifer, drained by a channel network down to the Northwest Branch, with the Wheaton Branch stormwater pond represented as a reservoir. Every input comes from public data sets, processed by scripts that are part of the project. It uses the same structure and method as the Rock Creek model and digital twin, and is calibrated against discharge and stage at USGS gage 01650800 (Sligo Creek near Takoma Park) over water years 2018–2019 and tested, without change, on 2020–2025. In both periods it reproduces hourly flow with a Nash–Sutcliffe efficiency of 0.7–0.8 and stage with 0.75–0.85. It over-predicts the total flow, by 14–16 % in the calibration years and 20 % over 2020–2025, mostly in the recessions after storms; about a third of the validation excess comes from the rainfall data set, which over-states the rain of 2022–2025. Over the same years it reproduces the gage much better than NOAA's National Water Model, above all at low flows.

## Contents

**1 What the twin does** **3**

**2 The watershed** **3**

**3 Model structure** **4**
3.1 Sub-catchments and channel network . . . . . 4
3.2 The hydrologic response unit . . . . . 7
3.3 Channels and gage section . . . . . 8
3.4 The Wheaton Branch stormwater pond . . . . . 9

**4 Governing equations** **11**

**5 Data** **13**
5.1 Discharge and stage . . . . . 14
5.2 Land cover and imperviousness . . . . . 15
5.3 Soils and terrain . . . . . 15
5.4 Meteorological forcing . . . . . 16

**6 Calibration** **17**
6.1 Parameter uncertainty . . . . . 20

**7 Validation** **24**
7.1 Comparison with the National Water Model . . . . . 26

**8 Limitations** **27**

**9 Software and reproducibility** **27**

# 1 What the twin does

Every six hours the twin

1. updates its rainfall with NOAA's Multi-Radar/Multi-Sensor (MRMS) radar–gauge precipitation analysis (hourly, 1 km, about one hour behind real time), scaled to the rainfall data set the model was calibrated with (Section 5.4), and its reference evapotranspiration with gridMET, followed by the most recent days of Open-Meteo;
2. advances the model from its last saved state to the present (the *analysis*);
3. runs the model seven days ahead from that state with the Open-Meteo hourly precipitation and reference-evapotranspiration forecasts (the *forecast*);
4. publishes the results: maps of each model element, time series of every variable in every sub-catchment, and model versus USGS real-time flow and stage at the gage.

The line shown for every variable is the model with the calibrated parameter set (Table 2). Alongside it the twin runs the same analysis and forecast with 200 parameter sets drawn from the posterior distribution of a Bayesian (Markov chain Monte Carlo) calibration; the 2.5 and 97.5 percentiles of these runs are the 95 % band that the viewer can shade. The band represents the uncertainty of the parameters only: errors of the rainfall, of the model structure and of the measurements are not included, and it is much narrower than the scatter of the measurements around the model. The model state is carried from cycle to cycle, so the twin keeps a continuous memory of soil moisture and groundwater. It does not assimilate observations: the model is not adjusted to match the gage, which makes the comparison with the gage shown in the viewer an ongoing, independent check.

# 2 The watershed

Sligo Creek rises in Wheaton, Maryland, and flows about 16 km south-east through Silver Spring and Takoma Park, mostly within a narrow stream-valley park along Sligo Creek Parkway, to the Northwest Branch at Hyattsville (Fig. 1). It drains 29.2 $km^2$: 84 % in Montgomery County, 13 % in Prince George's County and 3 % in Takoma, DC. The watershed is old, densely developed suburb served by storm drains throughout, and the stream has been the subject of extensive restoration and stormwater retrofits; the largest of these, the Wheaton Branch stormwater pond, is the only dam in the National Inventory of Dams. The part in the District drains to separate storm sewers that discharge to the creek, not to the DC combined sewer system. The USGS has recorded discharge and stage at Sligo Creek near Takoma Park (01650800, 16.7 $km^2$) since October 2008.

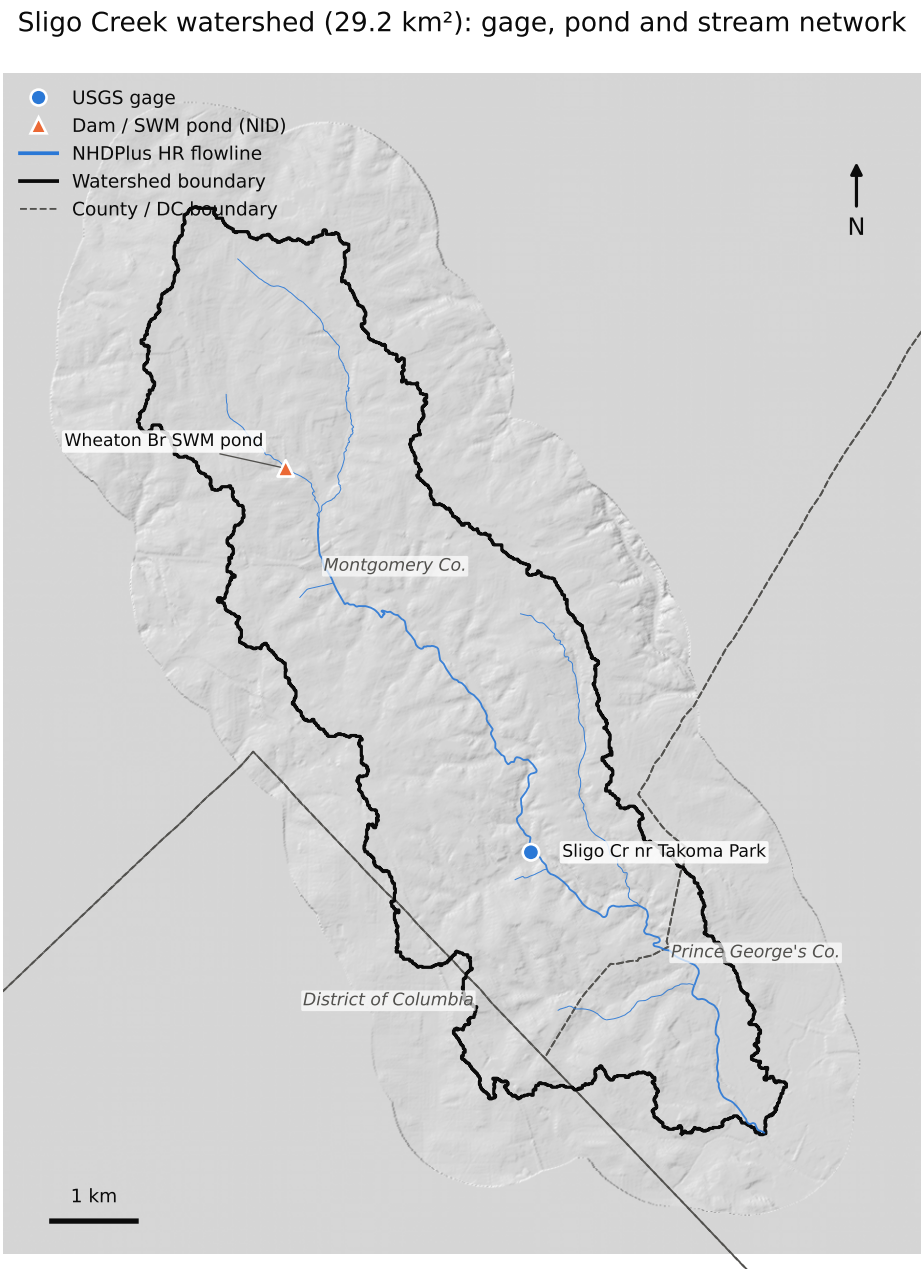



**Figure 1:** Sligo Creek watershed with the NHDPlus High Resolution stream network, the USGS gage, the Wheaton Branch stormwater pond (National Inventory of Dams) and the county and District boundaries.

# 3 Model structure

## 3.1 Sub-catchments and channel network

The watershed is delineated from the USGS 3DEP 10 m elevation model, with the NHDPlus High Resolution flowlines burned in (Fig. 2), into 27 sub-catchments of 0.5–2.1 $\mathrm{km}^2$ (median 0.9 $\mathrm{km}^2$) whose outlets are placed at the gage, the Wheaton Branch pond, the main confluences and the middle of the longest reaches (Fig. 3). Each sub-catchment drains to one channel segment; the

segments form the stream network down to the Northwest Branch. Bed elevations follow the smoothed NHDPlus HR profile, and the burned elevation model where the NHD has no flowline (Fig. 4).

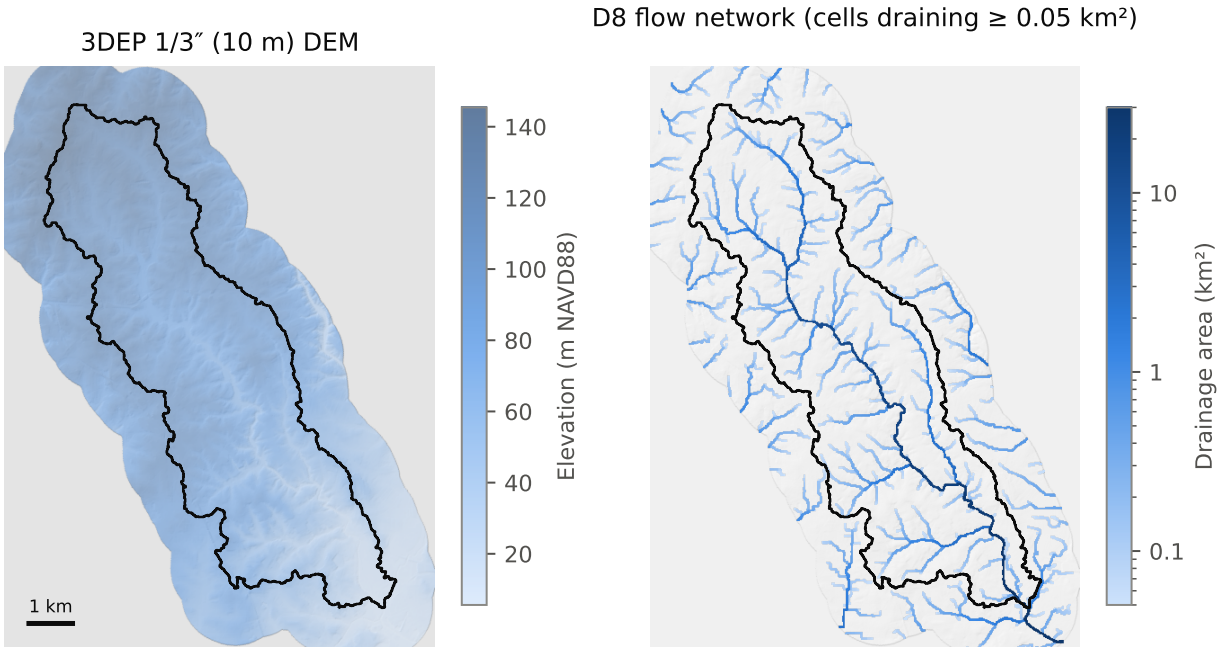



**Figure 2:** Left: 3DEP 10 m elevation model. Right: D8 flow network after burning in the NHD HR flowlines (cells draining at least 0.05 $km^2$).

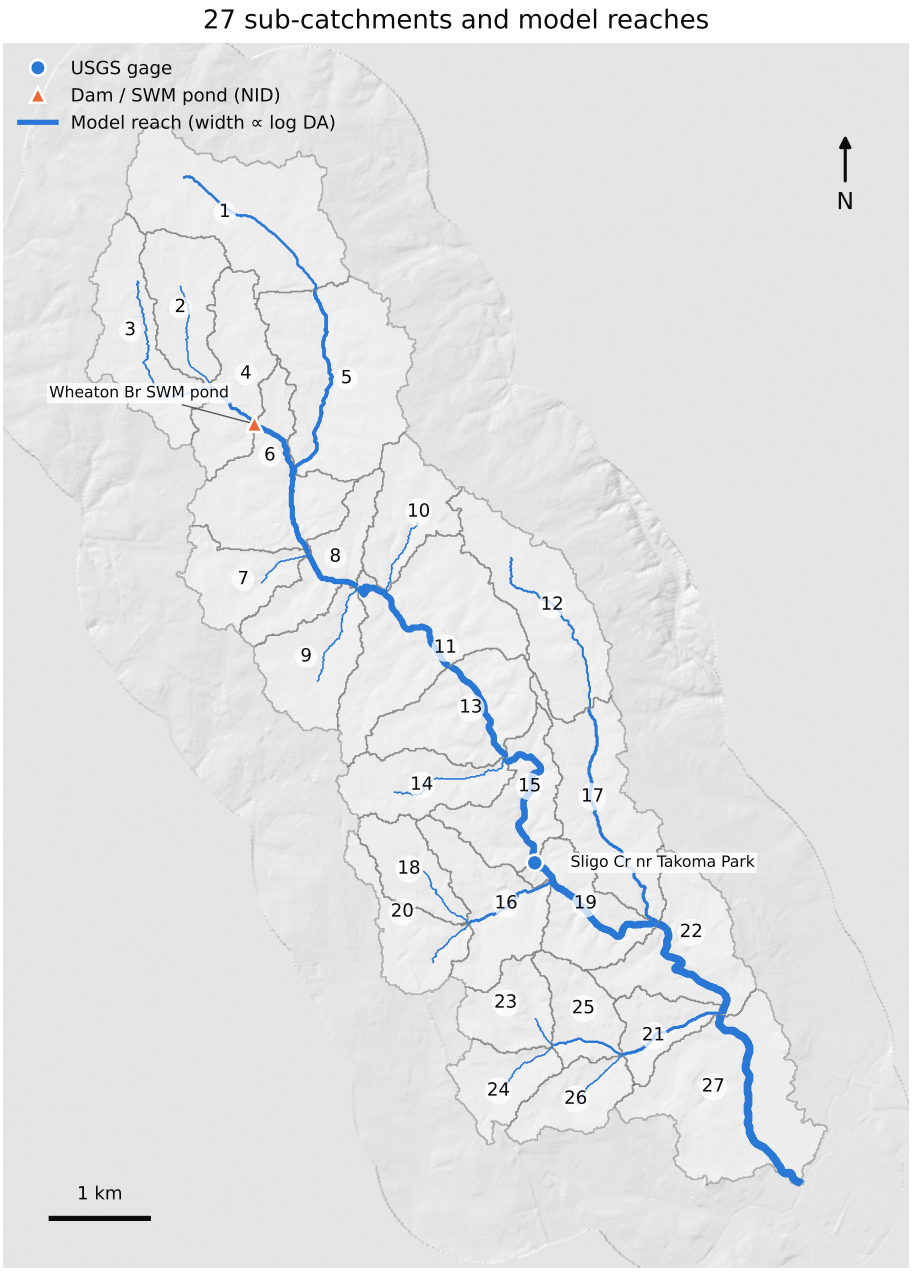



**Figure 3:** The 27 sub-catchments with their channel segments (line width proportional to the logarithm of drainage area) and control points.

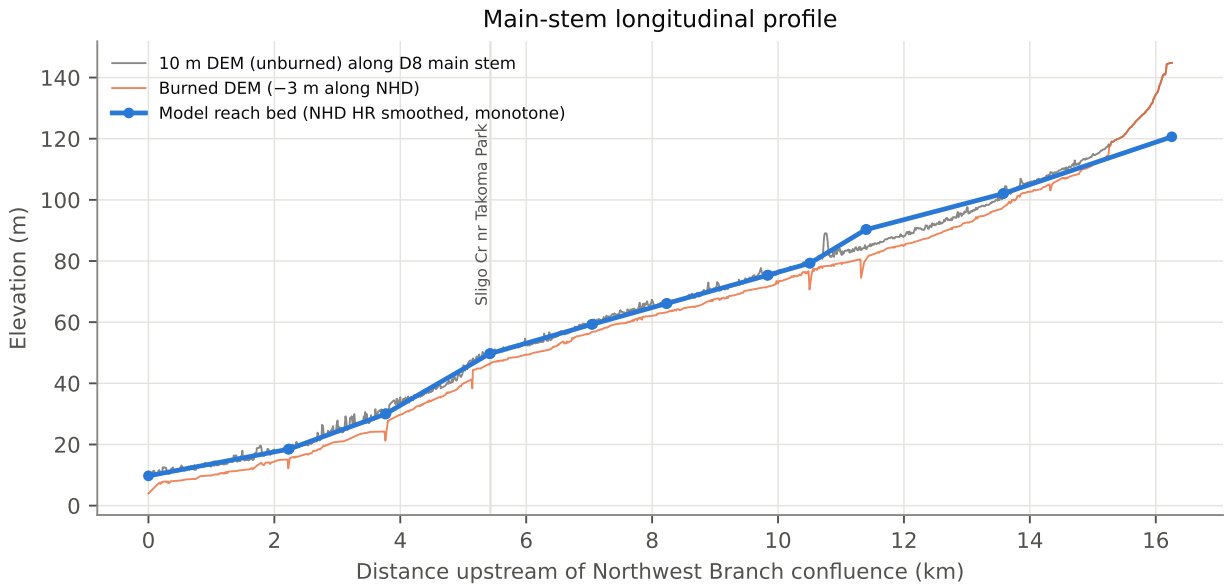



**Figure 4:** Main-stem longitudinal profile: elevation model along the main stem before and after burning, and the model bed elevations.

### 3.2 The hydrologic response unit

Each sub-catchment is one *Riparian Urban HRU*, a composite element of OHQ (Fig. 5):

- an **impervious surface** — the *directly connected* impervious area (roofs, roads and parking areas that drain to a storm sewer or the stream) — that drains through an internal impervious reach (gutters and storm pipes) to the stream;
- a **pervious surface** — lawns, forest and the impervious surfaces that drain onto them — that drains through an internal pervious reach (swales and small channels);
- three **soil layers** below the pervious surface, 0.5, 1 and 2 m thick, with their own hydraulic properties;
- a **regional aquifer** below the whole unit, exchanging groundwater with the neighbouring units;
- a **near-stream aquifer**, a strip along the stream that receives water from the regional aquifer and the soil above it and exchanges water with the stream in both directions.

The near-stream aquifer gives the unit a fast and a slow groundwater store in series: it fills from the stream and from recharge during storms and drains over days to weeks, reproducing the recessions observed after each event, while the regional aquifer supplies the slow seasonal baseflow.

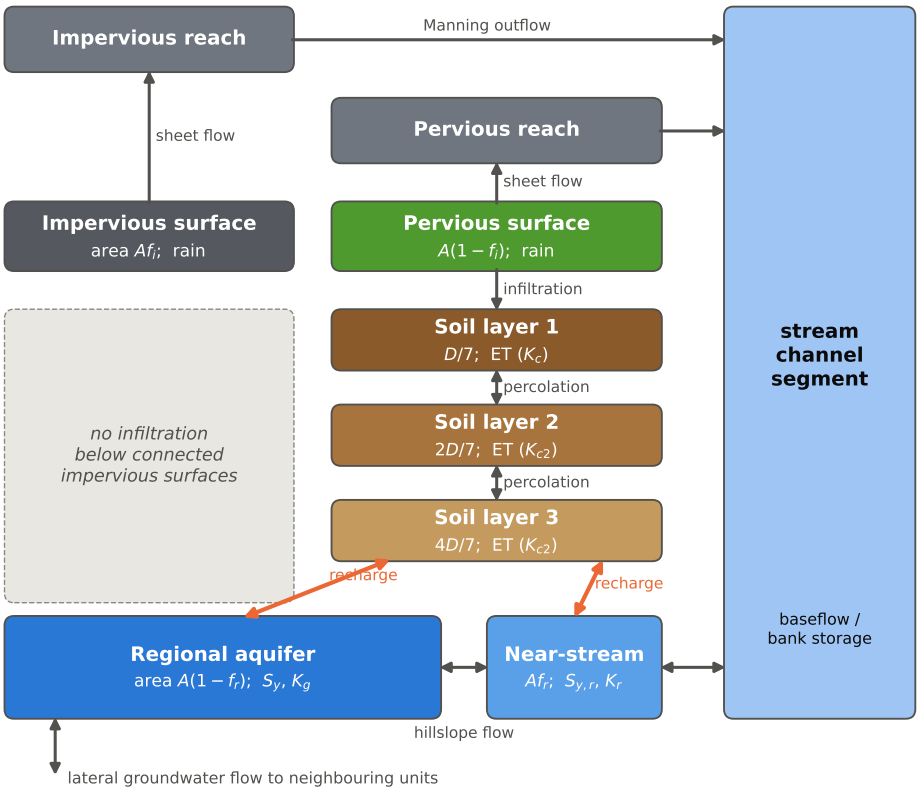



**Figure 5:** Structure of one sub-catchment (Riparian Urban HRU): storages (boxes) and fluxes (arrows; double arrows can flow both ways). $A$: area; $f_i$: connected impervious fraction; $f_r$: near-stream fraction; $D = 3.5\,\mathrm{m}$: soil depth.

## 3.3 Channels and gage section

Each channel segment has a compound cross-section: a power-law main channel ($A = aY^m$) below bank-full depth and a floodplain (flat width plus side slopes) above it. Away from the gage the main channel is scaled to the Maryland regional bank-full curves at the segment's drainage area: the Piedmont curves (McCandless and Everett, 2002) above the Fall Zone, and the Coastal Plain curves (McCandless, 2003) for the two lowest segments, which lie in the Coastal Plain according to the Maryland Geological Survey's physiographic map. Floodplain widths and side slopes come from valley cross-sections on the elevation model (Fig. 6). At the USGS gage a short gage segment carries a section fitted to the gage's continuous stage–discharge record; with it the model reproduces the observed stage for a given flow to within 1 % over the full range of flows (Fig. 7).

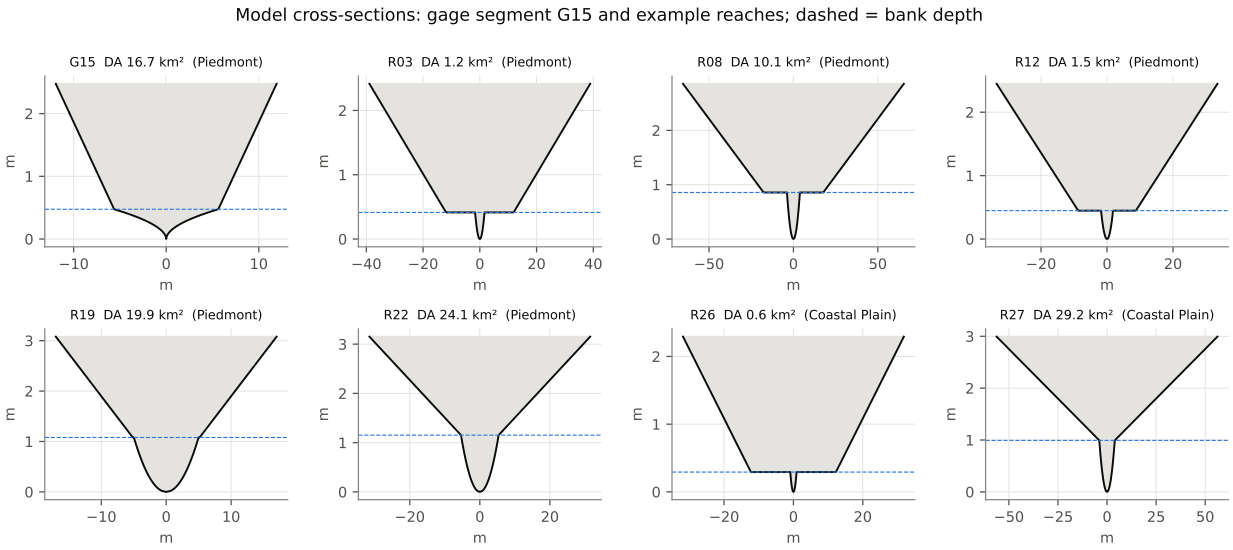



**Figure 6:** Model cross-sections: the gage segment and example channel segments.

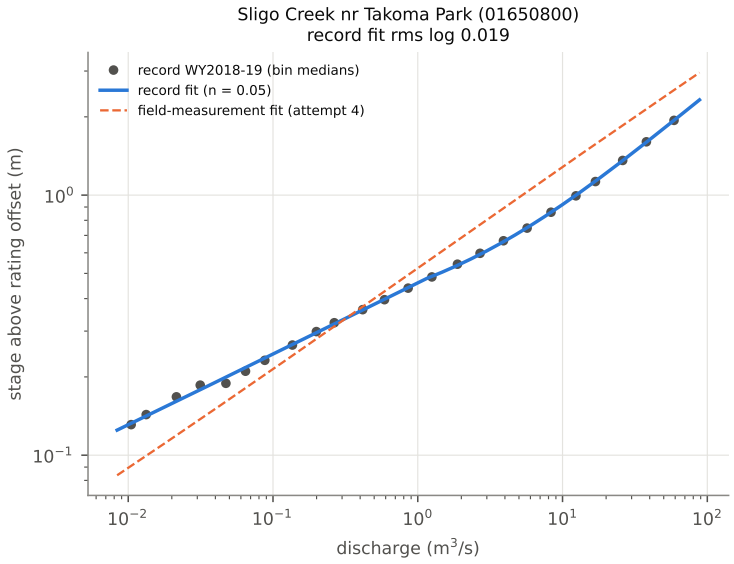



**Figure 7:** Gage 01650800: observed stage versus discharge (WY2018–19) and the fitted gage section.

## 3.4 The Wheaton Branch stormwater pond

The Wheaton Branch pond, a three-cell wet pond built for flood control in 1979 and retrofitted in 1988–90, captures the runoff of 3.1 km$^2$ of Wheaton. It is represented as a reservoir: its normal pool, the storage above it and the level of the embankment crest come from the 1 m lidar, and its outflow is a principal spillway (riser and conduit, with typical dimensions, since the design drawings are not public) plus overtopping of the armoured embankment crest (Fig. 8).

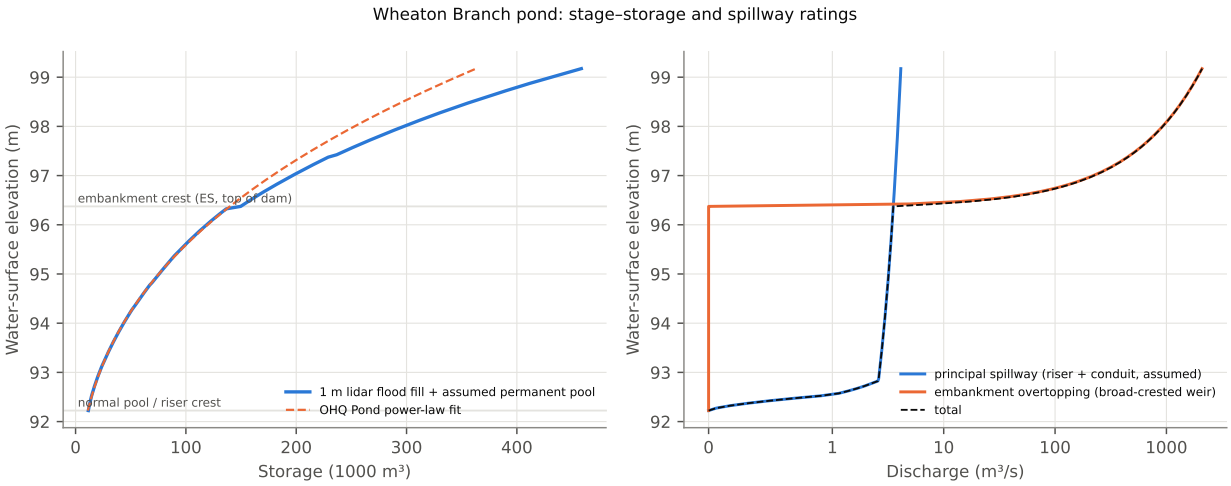



**Figure 8:** Wheaton Branch pond: stage–storage from the lidar and outflow ratings.

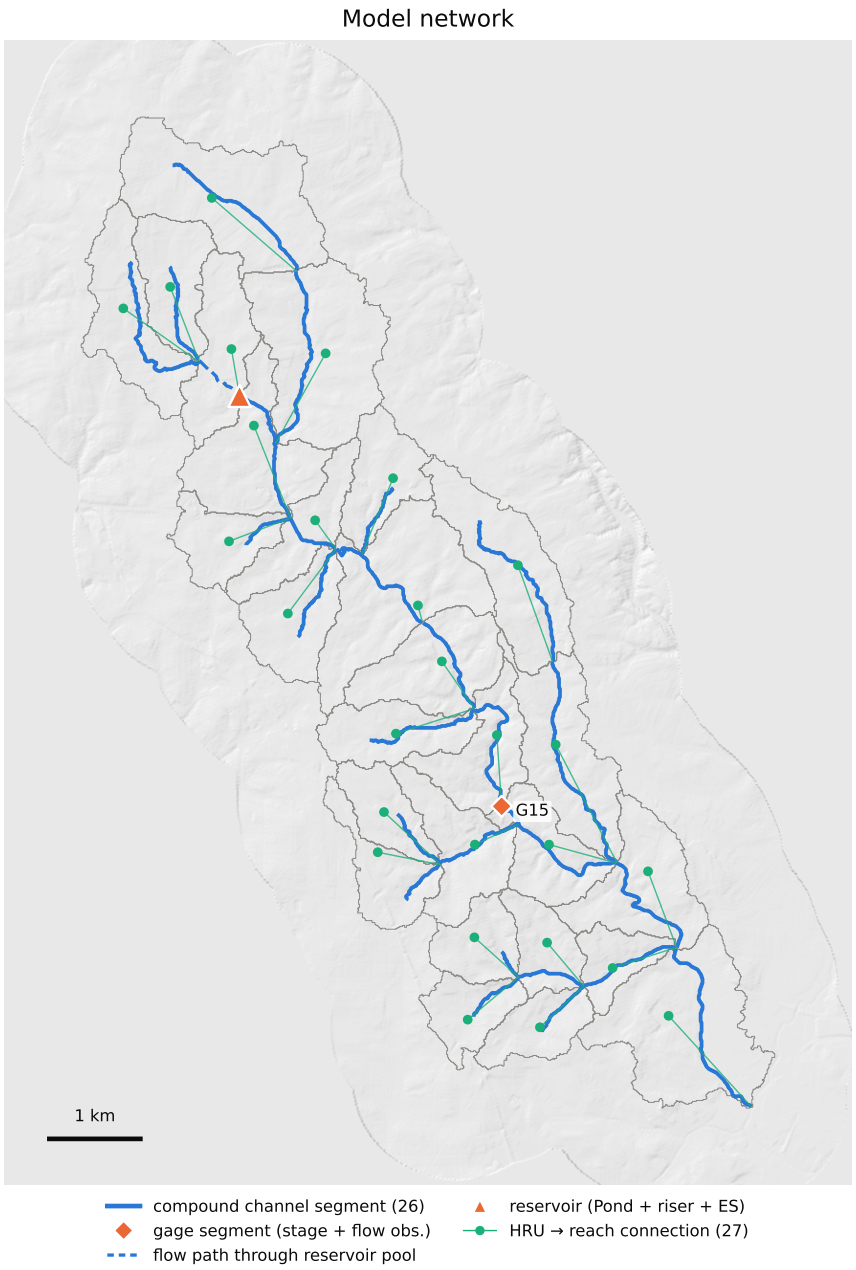



**Figure 9:** Model network: channel segments, gage segment, pond and the connection of each sub-catchment to its channel segment.

# 4 Governing equations

OpenHydroQual represents a system as a network of storage elements (blocks) connected by flux elements (links), each defined by user-editable expressions, and solves the resulting equations together (Massoudieh et al., 2023; the equations of the standard components are given in Massoudieh and Shakouri, 2025). It solves a water balance for every storage element (surface, reach, soil layer,

aquifer, channel segment, reservoir),

$$\frac{dV_j}{dt} = \sum_{k \in \mathrm{in}(j)} Q_k - \sum_{k \in \mathrm{out}(j)} Q_k + P_j - E_j, \tag{1}$$

where $V_j$ is the stored volume, $Q_k$ the flows of the connections (links) that enter and leave it, $P_j$ rainfall and $E_j$ evapotranspiration. The flows depend on the heads of the two elements they connect, as follows. Lengths are in metres and time in days; the Manning-type relations below give $\mathrm{m}^3/\mathrm{s}$ and are multiplied by 86 400.

**Surface runoff.** Water ponded on a surface above its depression storage $d_s$ runs off to the internal reach as sheet flow,

$$Q_s = \frac{W}{n_s} S^{1/2} \left(d - d_s\right)_+^{5/3}, \tag{2}$$

with overland-flow width $W$, slope $S$ and Manning coefficient $n_s$ ($(x)_+ = \max(x, 0)$).

**Internal reaches and channels.** Flow between a reach and the next segment, and between channel segments, follows Manning's equation driven by the difference in water-surface elevation,

$$Q = \frac{1}{\bar{n}} A_u R^{2/3} \left(\frac{h_1 - h_2}{\bar{L}}\right)^{1/2}, \tag{3}$$

with the flow area $A_u$ of the upstream end, hydraulic radius $R$, mean length $\bar{L}$ and mean roughness $\bar{n}$; backwater and flow reversal are therefore represented.

**Soil water.** Each soil layer follows the van Genuchten–Mualem relations,

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha}\left(S_e^{-1/m} - 1\right)^{1/n}, \quad k_r = \left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2, \quad m = 1 - \tfrac{1}{n}, \tag{4}$$

with total head $h = z + \psi$. *Infiltration* into the top layer is Darcy flow over half its thickness, $Q = A\,K_{s,1}\,(h_s - h_1)/(d_1/2)$, and *percolation* between layers is $Q = A\,K_s\,k_r(S_e)\,(h_i - h_j)/\bar{d}$, upward when the lower layer has the higher head (capillary rise).

**Evapotranspiration.** Each soil layer loses

$$E = K_c\,E_0(t)\,A\,\frac{\theta}{\theta_s}, \tag{5}$$

where $E_0$ is the reference (grass) evapotranspiration and $K_c$ a crop coefficient, one for the top layer and one for the two deeper layers.

**Recharge.** The bottom soil layer drains to each aquifer through its unsaturated conductivity, and receives water from it at the saturated conductivity when the water table is higher,

$$Q_r = A_r\,K_{s,3}\,\frac{k_r(S_{e,3})\,(h_3 - h_g)_+ - (h_g - h_3)_+}{L}, \tag{6}$$

over the share $A_r$ of the pervious area above that aquifer.

**Aquifers.** The water table of each aquifer cell follows from its storage and specific yield $S_y$, $h_g = z_b + V/(S_y A)$. Groundwater moves between neighbouring units, and from the regional to the near-stream aquifer, by Darcy's law with the saturated thickness $b$ as transmissive thickness,

$$Q = \bar{K}\,\bar{b}\,W\,\frac{h_1 - h_2}{L}, \tag{7}$$

with interface width $W$ (the shared boundary between units; twice the stream length within a unit for the regional–near-stream exchange) and length $L$ (centroid distance; the hillslope length for the regional–near-stream exchange). Lateral outflow tapers to zero as a cell approaches the depth of its stream bed, so that one unit cannot be drained by its neighbours.

**Baseflow and bank storage.** The near-stream aquifer exchanges water with the channel,

$$Q_b = K_r\,A_b\,\frac{h_g - h_c}{L_b}, \tag{8}$$

with bank area $A_b$ and path length $L_b$; the flow reverses when the stream is above the water table.

**Wheaton Branch pond.** The stage of the pond follows from its storage through the fitted stage–storage relation $V = \alpha(h - z_b)^{\beta}$. The principal spillway discharges the lesser of the riser weir and the conduit capacity, $Q = \min\!\big(C_w L H^{3/2},\ \tfrac{\pi}{4} D^2 \sqrt{2g\,\Delta h/(1+K)}\big)$; overtopping of the embankment crest is a broad-crested weir, $Q = C\,L\,(h - z_c)_+^{3/2}$.

**Numerical solution.** The equations of all 271 storages are solved together with an implicit (backward Euler) scheme and Newton iterations, with an adaptive time step of at most 3.6 min (five times the initial step of 43 s). Longer steps made the likelihood jump between almost identical parameter sets, because whether the solver cuts a step depends on tiny differences; at this limit the likelihood changes by less than 0.2 for a $10^{-6}$ relative change of all parameters. The flow of each step is the average over the step and is placed at its midpoint when the output is written. For speed the model is compiled into C++ by the OHQ code generator, which reproduces the interpreter's solution; one year of simulation takes about a minute on one processor core.

# 5 Data

All inputs are public. Table 1 lists the sources; the maps below show how they enter the model.

**Table 1:** Data sources.

| Model input | Source |
|---|---|
| Terrain, drainage network | USGS 3DEP 10 m DEM and 1 m lidar; NHDPlus High Resolution |
| Land cover, imperviousness | Chesapeake Conservancy 1 m land use/land cover (2024 edition) |
| Connected impervious area | Sutherland (2000) relation, checked against public storm-drain inventories |
| Soils | NRCS SSURGO; van Genuchten parameters from Rosetta3 |
| Channels | Maryland Piedmont and Coastal Plain bank-full relations; MGS physiographic provinces; USGS continuous stage–discharge record |
| Pond | National Inventory of Dams; USGS 3DEP 1 m lidar |
| Combined-sewer check | DC GIS, combined sewer system area and MS4 sewersheds |
| Rainfall (calibration, validation) | NOAA Analysis of Record for Calibration (AORC), hourly, 1 km |
| Rainfall (operational) | NOAA MRMS multi-sensor QPE (hourly, 1 km); Open-Meteo forecast |
| Reference ET | gridMET (daily, 4 km); Open-Meteo FAO-56 reference ET (recent days and forecast) |
| Discharge and stage | USGS National Water Information System |

## 5.1 Discharge and stage

Gage 01650800 records discharge and stage every 5 min; the record is 94–100 % complete in every water year since 2009 except WY2011 (Fig. 10). The Northwest Branch gage near Hyattsville (01651000), which includes Sligo Creek, was discontinued in November 2023 and serves only as a historical check.

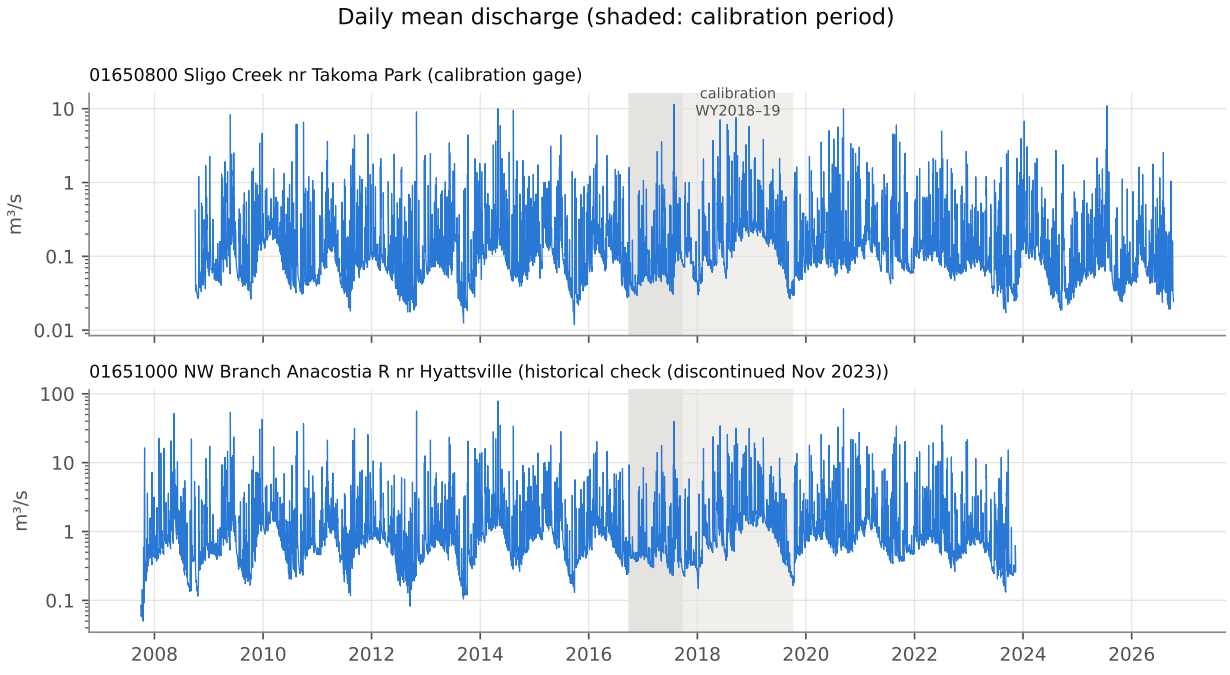



**Figure 10:** Daily mean discharge at Sligo Creek near Takoma Park and at the Northwest Branch near Hyattsville. Shading: calibration period (dark) and model spin-up (light).

## 5.2 Land cover and imperviousness

Land cover comes from the Chesapeake Bay 1 m land use/land cover data set (2021 imagery). Sligo Creek is 41 % impervious (roads, buildings, parking and other paved surfaces, including those under tree canopy), ranging from 24 % to 61 % among the sub-catchments (Fig. 11). Only the impervious area that drains directly to a storm drain or the stream, the directly connected impervious area (DCIA), produces fast storm runoff in the model; the rest drains onto lawns and soil. DCIA is estimated from the total impervious area with the "average" relation of Sutherland (2000), $\mathrm{DCIA} = 0.1\,\mathrm{TIA}^{1.5}$ (percent), giving 27 % of the watershed; one watershed-wide multiplier on DCIA is calibrated. An independent estimate from the public storm-drain inlet inventories (Montgomery County, Maryland State Highway Administration, City of Takoma Park, DC Water) has the same spatial pattern and is about 30 % lower, as expected for an inventory that lacks private drains.

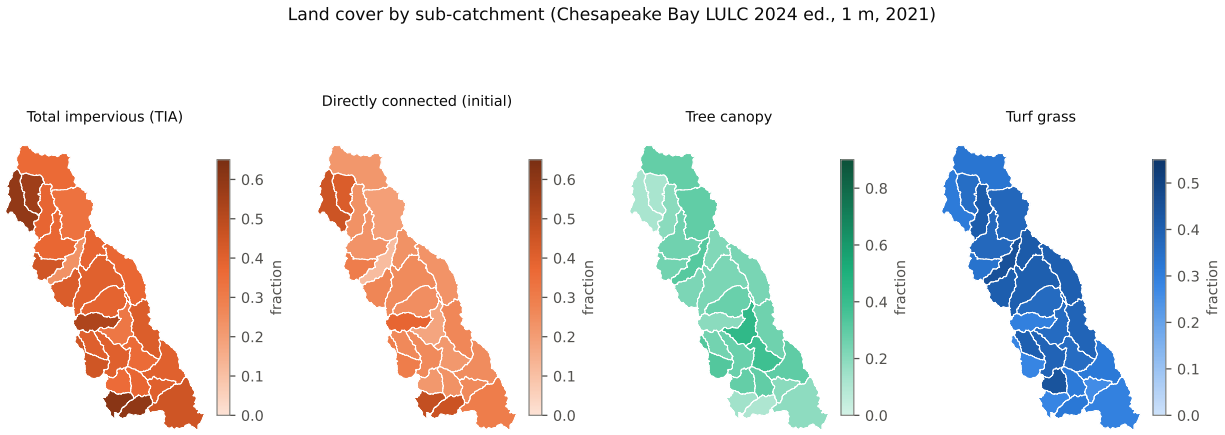



**Figure 11:** Total and directly connected impervious fraction, tree canopy and turf grass by sub-catchment.

## 5.3 Soils and terrain

Soil hydraulic properties for the three soil layers of each sub-catchment come from the NRCS SSURGO database: saturated conductivity from the survey, and the water-retention parameters from the Rosetta3 pedotransfer model. Three quarters of the watershed is in hydrologic soil group B; the slower C and D soils are mostly in the lowest sub-catchments in Prince George's County (Fig. 12). Overland-flow length and width, slopes and the length of each sub-catchment's internal drainage network come from the 10 m elevation model (Fig. 13).

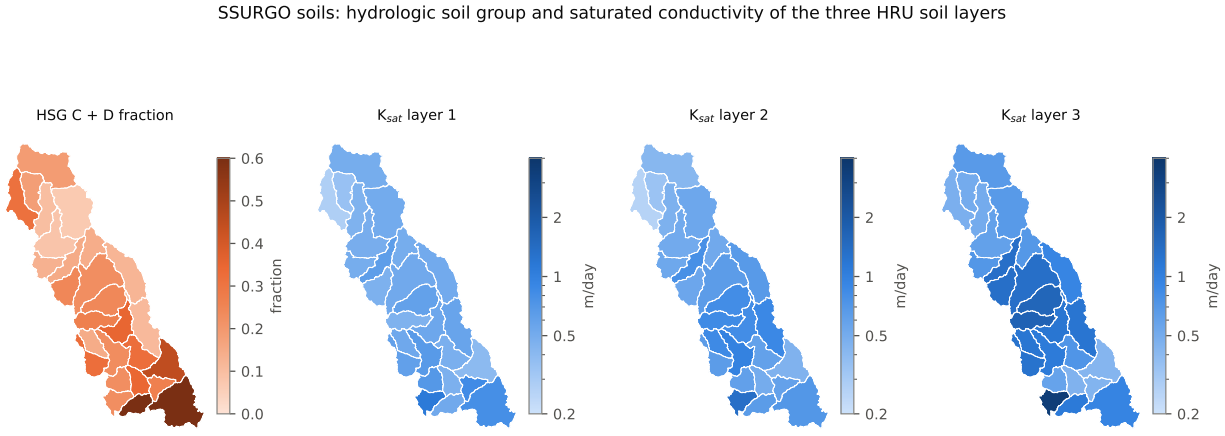



**Figure 12:** Fraction of hydrologic soil groups C and D, and saturated conductivity of the three soil layers.

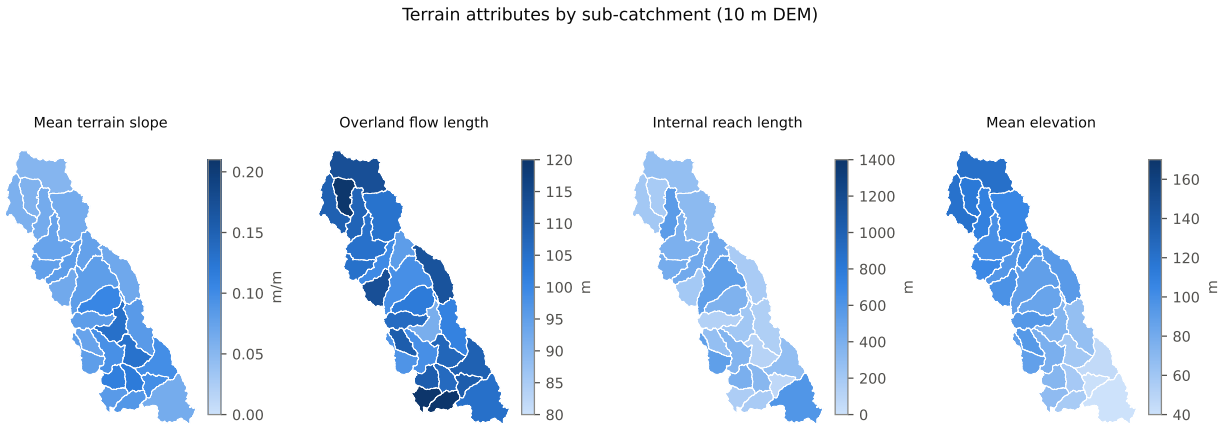



**Figure 13:** Terrain attributes by sub-catchment.

## 5.4 Meteorological forcing

For calibration and validation the model is driven by NOAA's Analysis of Record for Calibration (AORC), hourly precipitation on a 1 km grid, and by gridMET daily reference evapotranspiration (4 km), both averaged over each sub-catchment. The watershed receives about 1220 mm of precipitation a year (1015–1452 mm by water year); reference ET is about 1180 mm/yr (Fig. 14). AORC agrees with the Reagan National Airport gauge to within 8 % (daily correlation 0.96). The twin uses NOAA's real-time Multi-Radar/Multi-Sensor (MRMS) precipitation. Over 15 October 2020 – 30 September 2025 (1811 days) the two agree closely in timing (daily correlation 0.99, 96 % agreement on wet and dry days), but AORC is 16.5 % wetter in total (11–24 % by season, 10–25 % by sub-catchment, 12 % on days with more than 25 mm). MRMS is therefore multiplied by 1.165 to stay consistent with the rainfall the parameters were calibrated on; the twin inherits the model's tendency to over-predict flow with AORC (Section 7). gridMET and the Open-Meteo reference ET agree to within about 4 % over their overlap.

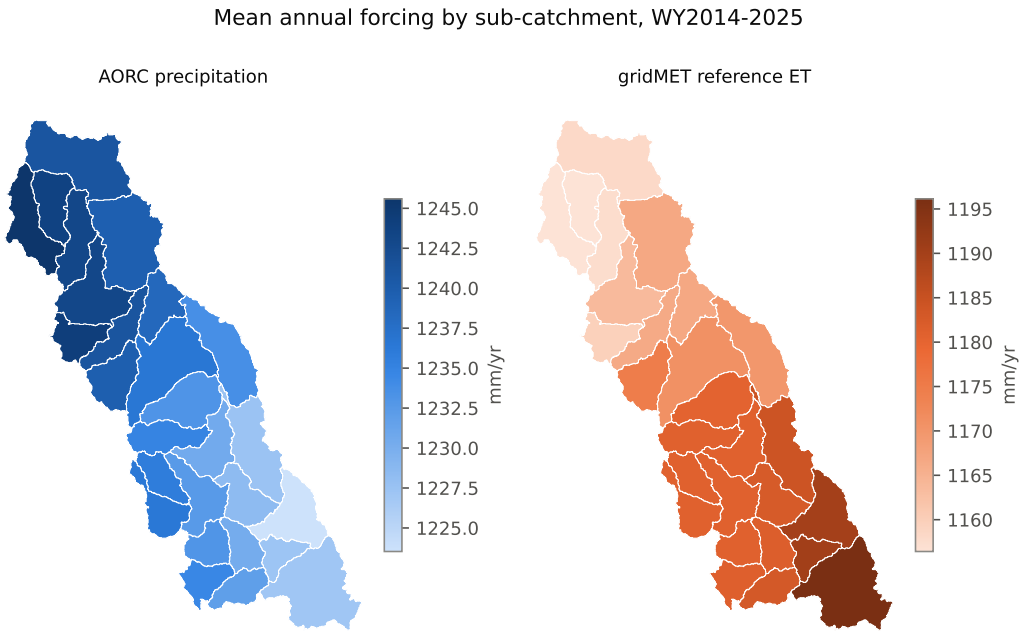



**Figure 14:** Mean annual AORC precipitation and gridMET reference ET by sub-catchment, WY2014–2025.

# 6 Calibration

Twelve parameters were estimated, each applied identically to all 27 sub-catchments (Table 2), so that the differences between sub-catchments come entirely from the data above. They were fitted to discharge and stage at USGS 01650800 over water years 2018–2019 (hourly values during storms, daily values otherwise; the model was run from October 2013, so the first four years serve as spin-up). The objective is the likelihood of the log-transformed observations with first-order autoregressive (AR(1)) errors: each series has its own error standard deviation and correlation time, both estimated, so that consecutive hourly residuals during a storm are not counted as independent information (the estimated correlation times are 3.0 h for flow and 2.8 h for stage). A genetic algorithm, started from the final Rock Creek parameters, explored the parameter space; a Bayesian (Markov chain Monte Carlo) sampling of the posterior distribution followed, and the final set (Table 2) is the Levenberg–Marquardt optimum started from the best sample of that run; the posterior uncertainty is in Section 6.1.

**Table 2:** Calibrated parameters (final values) with their search ranges and, for comparison, the Rock Creek values.

| Parameter | Role | Range | Sligo Creek | Rock Creek |
|---|---|---|---|---|
| Soil conductivity multiplier | scales SSURGO $K_s$ of all layers | 0.001–100 | 20.2 | 4.42 |
| Surface conductivity multiplier | additional factor on the top layer (compaction) | 0.001–10 | 0.0010 | 0.019 |
| Connected-impervious multiplier | scales the Sutherland DCIA | 0.3–3 | 0.42 | 1.22 |
| Crop coefficient, top layer | $K_c$ | 0.3–1.4 | 0.36 | 0.67 |
| Crop coefficient, deeper layers | $K_{c2}$ | 0–1 | 0.26 | 0.29 |
| Regional aquifer conductivity | m/d | 0.01–100 | 27 | 0.023 |
| Regional aquifer specific yield | – | 0.06–0.4 | 0.27 | 0.060 |
| Near-stream fraction | $f_r$ | 0.005–1 | 0.45 | 0.054 |
| Near-stream specific yield | – | 0.005–0.4 | 0.013 | 0.39 |
| Near-stream and bank conductivity | m/d | 0.01–100 | 13 | 1.02 |
| Channel roughness | Manning $n$ | 0.02–0.1 | 0.051 | 0.050 |
| Pond / reservoir riser rating multiplier | – | 0.02–3 | 0.12 | 0.84 |

Three results stand out. First, the connected impervious area is 0.42 times the Sutherland estimate, about 11 % of the watershed (Rock Creek: 21 %); the storm-drain inventories, which lack private drains, gave 19 %. Second, the top soil layer conducts about 2 % of the SSURGO value (surface multiplier times soil multiplier), consistent with compacted urban soils, while the deeper layers conduct twenty times the SSURGO value: the soil below the surface drains much faster than the survey's values, which may stand for flow paths the model does not have, such as utility and pipe trenches. Third, the groundwater is fast: a conductive regional aquifer drains into a near-stream zone that covers almost half of each sub-catchment and has a small specific yield, so it rises and drains within days. Together these move part of the storm response from connected pavement to a quick subsurface path, and less water is lost to evapotranspiration than with the Rock Creek values. One gage constrains only combinations of these parameters: the Bayesian sampling finds two groups of nearly equally good sets, which differ mainly in the outflow of the Wheaton Branch pond (in the other group the riser rating is about 0.6 times that of Table 2).

Over the calibration years the model reproduces hourly flow with a Nash–Sutcliffe efficiency of 0.71 in both water years (0.78 and 0.88 on the logarithms) and stage with 0.81 and 0.86 (Fig. 15). It over-predicts the total flow by 14–16 %. The exceedance curves (Fig. 16) show where: the model is within 10 % of the measured flow from 5 to 95 % exceedance except in the transition from storm flow to baseflow (about 25 % exceedance, 19 % high), 10–16 % high in the driest hours, and up to 54 % high for moderate storm flows of 3–6 $m^3/s$ (about 1 % of the hours); the largest flows are within 15 %. Stage is within 7 % except for the highest 1 % of the hours, where it is about 13 % high.

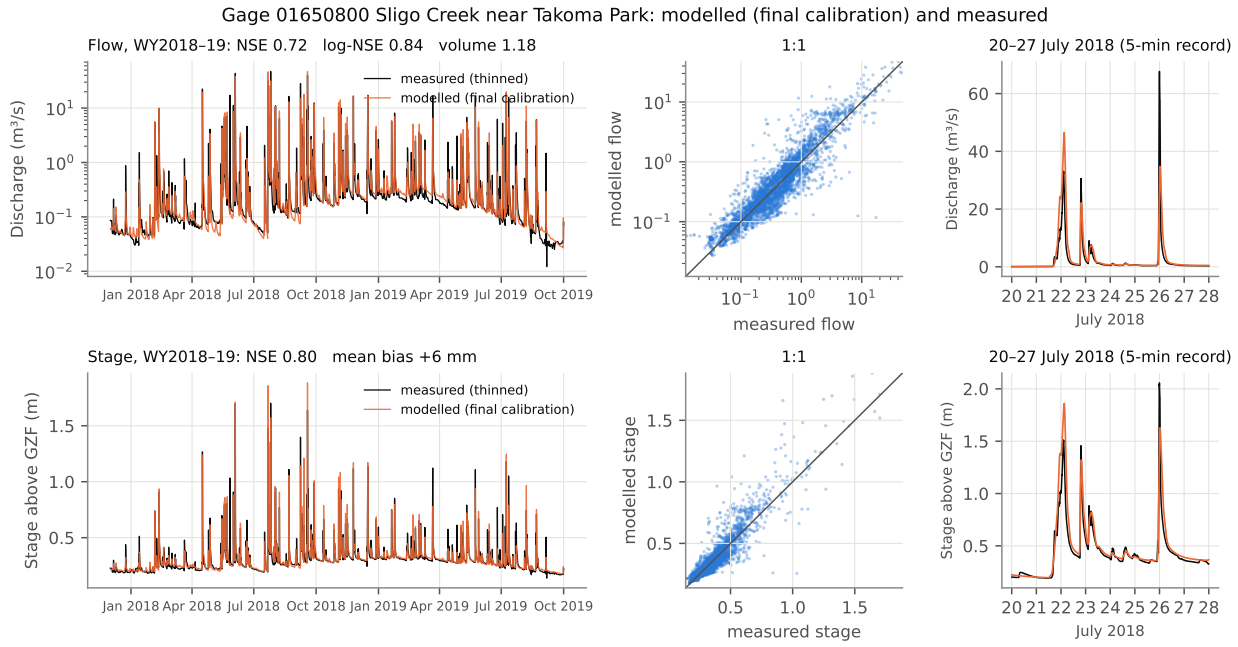



**Figure 15:** Calibration years: measured and modelled flow and stage at the USGS gage, 1:1 plots, and two storms in July 2018 against the 5-minute record.

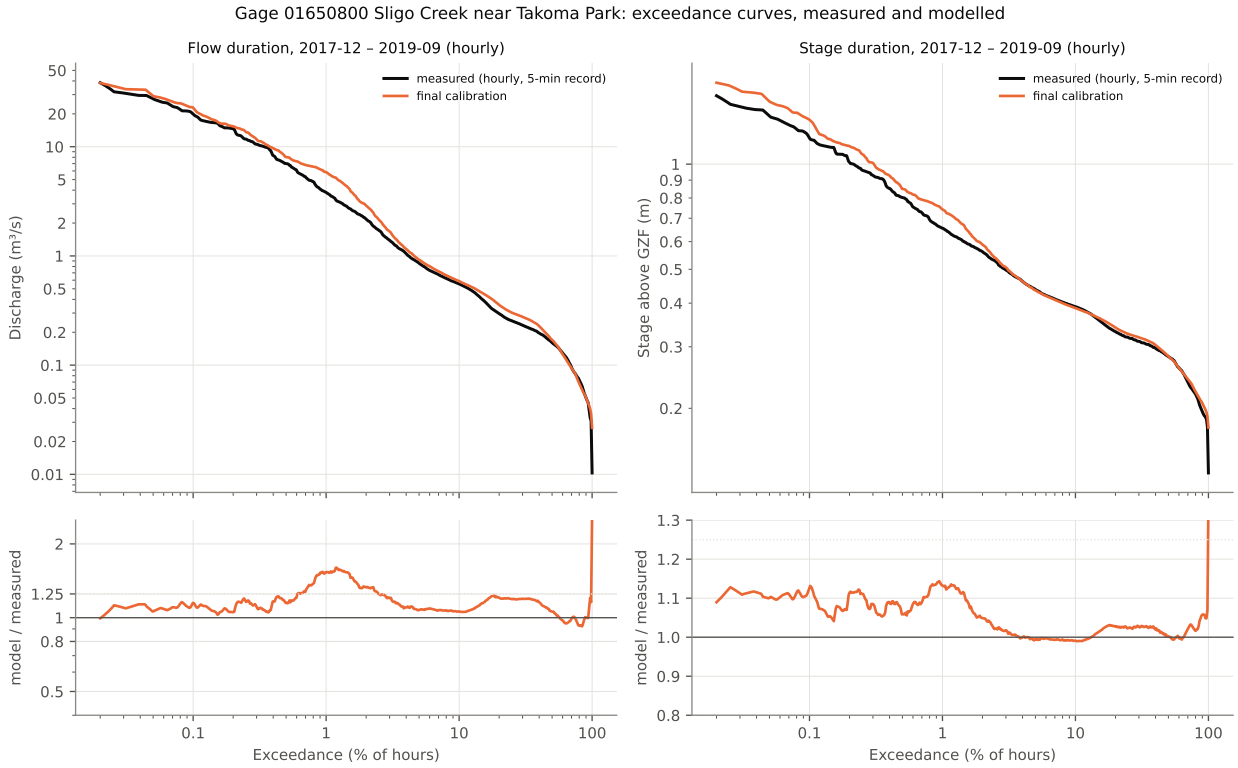



**Figure 16:** Calibration years: hourly flow and stage exceedance (duration) curves, measured and modelled, December 2017 – September 2019 (log–log), and the ratio of modelled to measured values at equal exceedance.

## 6.1 Parameter uncertainty

The uncertainty of the parameters was estimated by sampling their posterior distribution with a Markov chain Monte Carlo method (adaptive Metropolis with a proposal shaped by the covariance of the samples), with the same AR(1) likelihood and log-normal priors whose ±2 standard deviations are the ranges of Table 2. Eight chains were started around the calibrated set and run for 800 steps each; the statistics use the last 350 steps (2792 samples). The chains agree well for most parameters (split-$\hat{R}$ 1.07–1.26) and less well for the crop coefficients and the regional specific yield (1.35–1.57), which can partly compensate each other, so the intervals of these three are approximate. The posterior is narrow (Table 3 and Fig. 17): the 95 % intervals extend from 3–5 % either side of the median (channel roughness, error standard deviations) to 15–30 % (soil, groundwater and pond parameters), and the calibrated set lies inside every interval except that of the regional aquifer conductivity (27.3 against 18.2–26.9 m/d), where the samples favour a slightly slower aquifer.

**Table 3:** Posterior distribution of the parameters: median and 95 % interval, and the calibrated (Levenberg–Marquardt) value.

| Parameter | Calibrated | Median | 95 % interval |
|---|---:|---:|---:|
| Soil conductivity multiplier | 20.2 | 20.7 | 17.0–24.7 |
| Surface conductivity multiplier | 0.0010 | 0.0010 | 0.0008–0.0013 |
| Connected-impervious multiplier | 0.42 | 0.41 | 0.38–0.44 |
| Crop coefficient, top layer | 0.36 | 0.34 | 0.29–0.38 |
| Crop coefficient, deeper layers | 0.26 | 0.28 | 0.25–0.31 |
| Regional aquifer conductivity (m/d) | 27.3 | 22.5 | 18.2–26.9 |
| Regional aquifer specific yield | 0.27 | 0.27 | 0.22–0.30 |
| Near-stream fraction | 0.45 | 0.46 | 0.43–0.50 |
| Near-stream specific yield | 0.013 | 0.012 | 0.010–0.014 |
| Near-stream and bank conductivity (m/d) | 13 | 12 | 10–13 |
| Channel roughness (Manning $n$) | 0.051 | 0.050 | 0.049–0.053 |
| Pond riser rating multiplier | 0.12 | 0.12 | 0.11–0.15 |
| Error s.d., log flow | 0.45 | 0.45 | 0.43–0.46 |
| Error s.d., log stage | 0.12 | 0.13 | 0.12–0.13 |

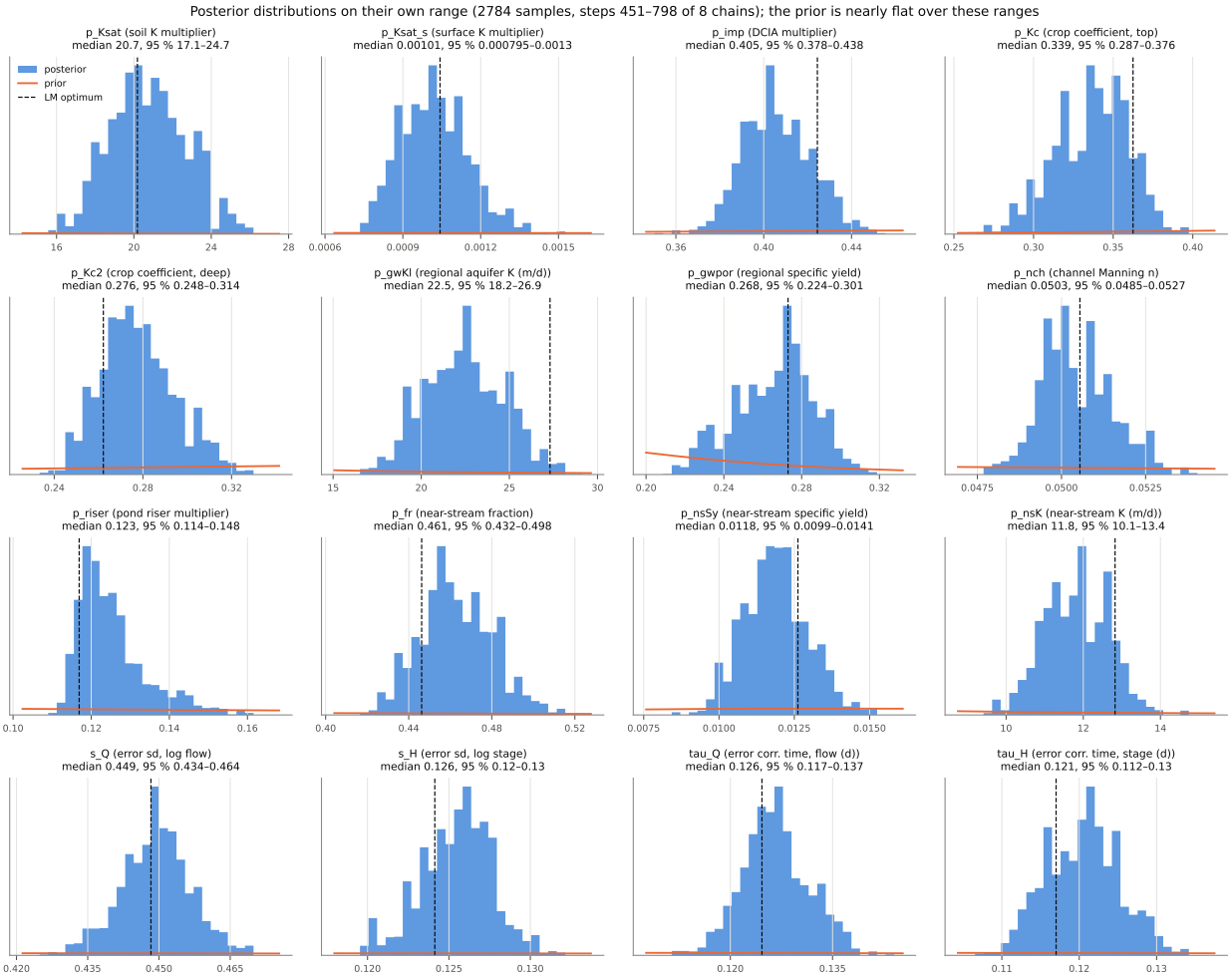



**Figure 17:** Posterior distributions of the parameters (histograms), the prior density over the same ranges (nearly flat) and the calibrated value (dashed).

Running eighty posterior parameter sets through the model shows what this means for the simulated flow and stage (Figs. 18 to 20). The 95 % credible interval of the model output, the band the twin shows, is narrow: its median width is 10 % of the flow and 2 % of the stage, and it contains only 14 % of the measured hours. Adding the error model of the calibration (log-normal, standard deviation 0.45 for flow and 0.13 for stage) gives a 95 % prediction interval that contains 97.6 % of the measured hours, as it should, but is wide (about a factor of 2.4 either side of the flow). Nearly all of the difference between model and measurements is therefore not parameter uncertainty but error of the model structure and the rainfall: the systematic over-prediction of moderate storms and recessions lies outside the parameter band at most exceedance levels.

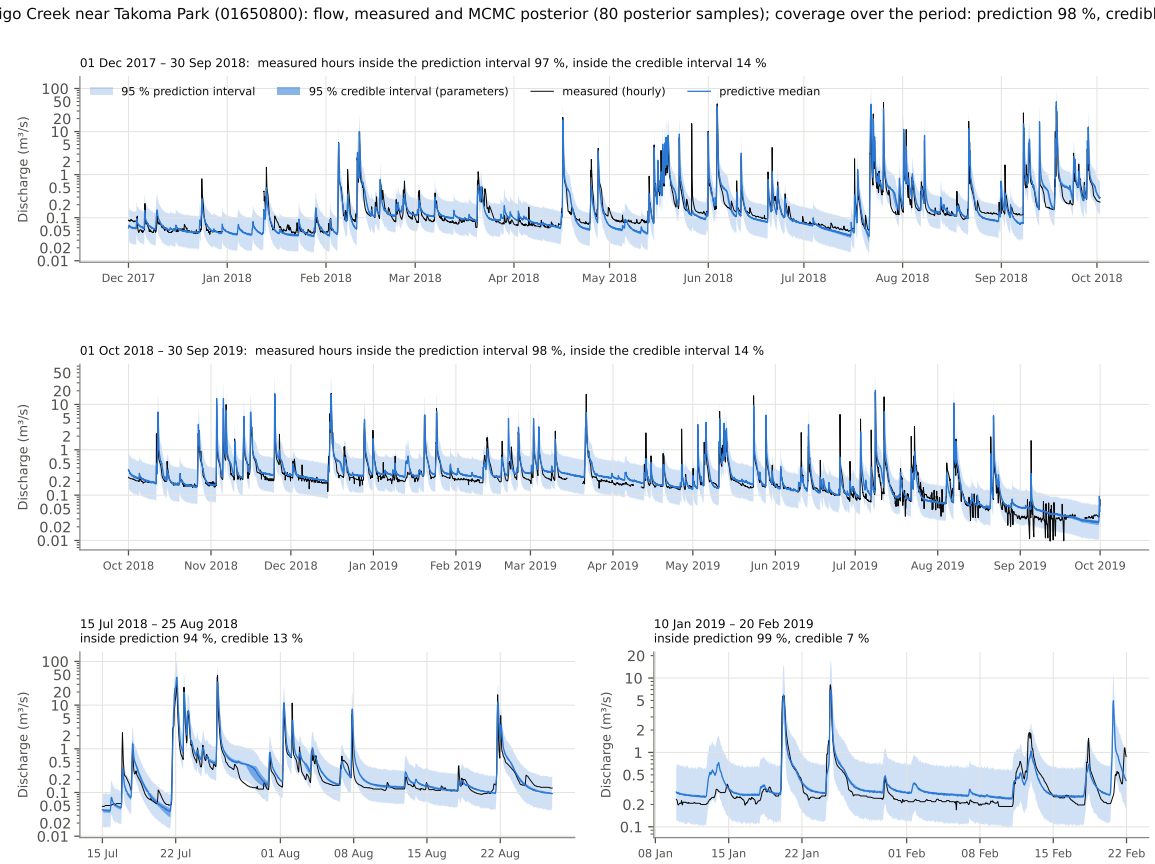



**Figure 18:** Calibration years: measured hourly flow with the 95 % credible interval of the model (parameter uncertainty, 80 posterior samples) and the 95 % prediction interval (with the calibrated log-normal error), two years and two six-week windows.

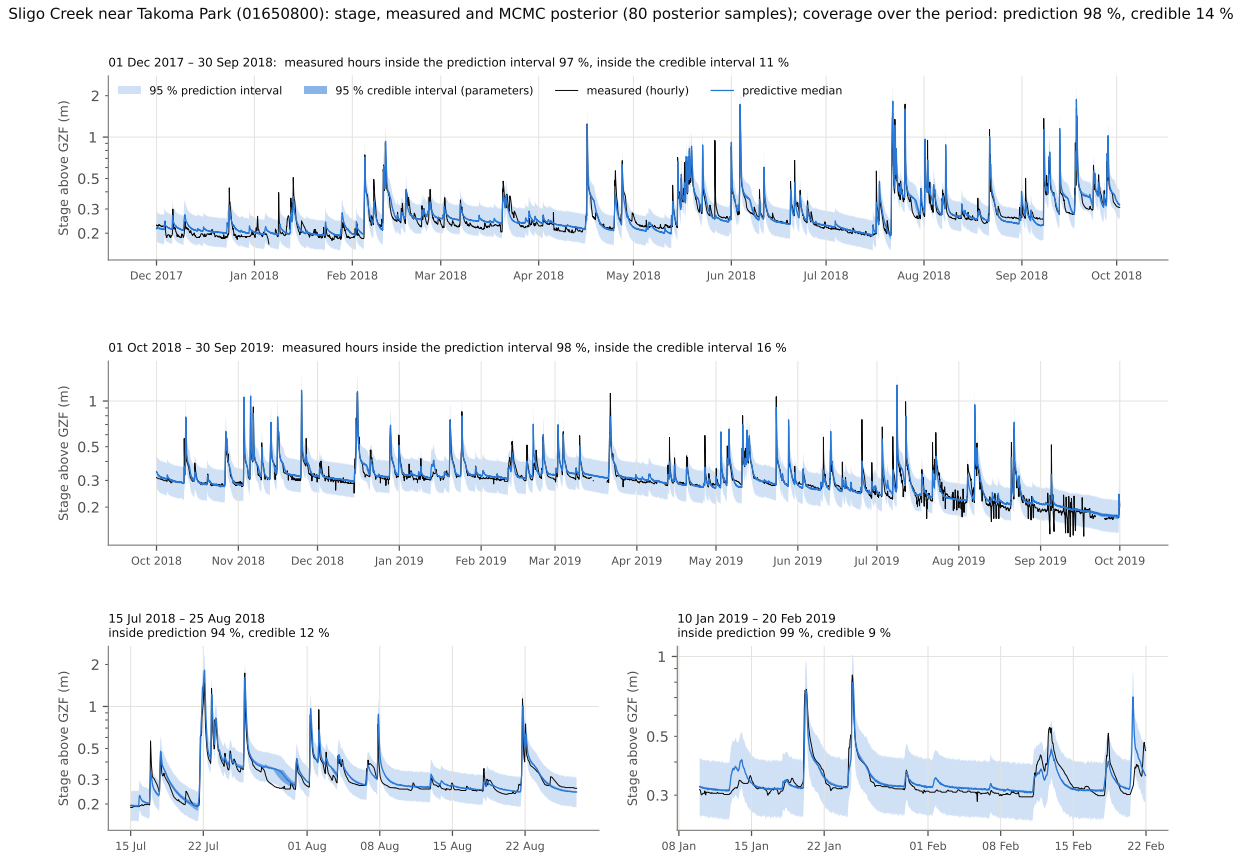



**Figure 19:** As Fig. 18, for stage.

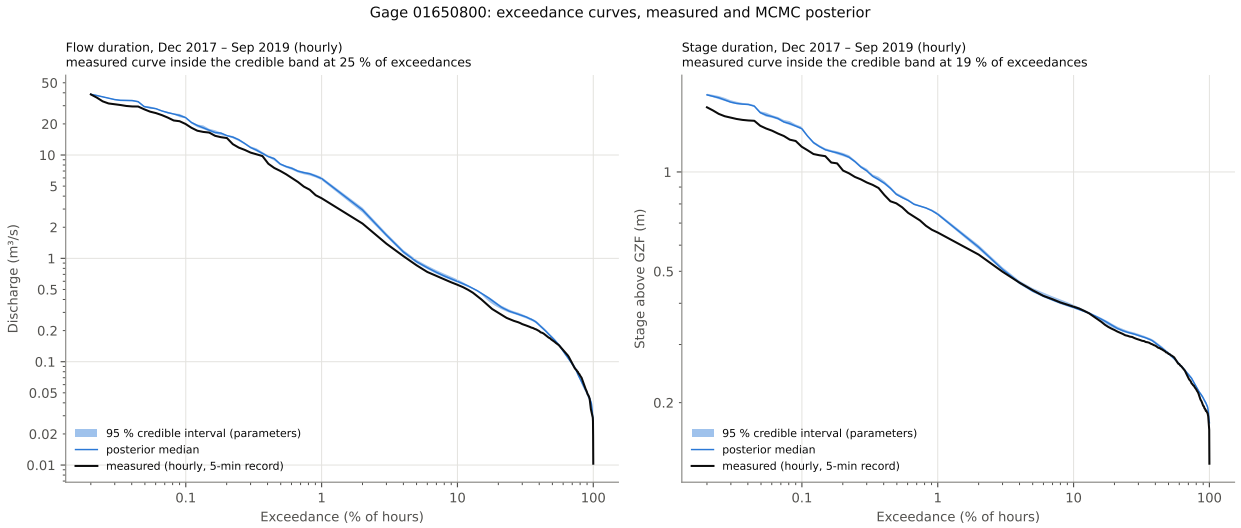



**Figure 20:** Calibration years: hourly flow and stage exceedance curves, measured and the posterior median with its 95 % credible interval.

# 7 Validation

The calibrated model was run without any change from October 2013 to September 2025 and compared with the USGS record of water years 2020–2025, which was not used in the calibration (Tables 4 and 5 and Figs. 21 and 22). It reproduces the timing and size of storms about as well as in the calibration years: hourly flow Nash–Sutcliffe efficiency 0.43–0.87 by year (0.75 at the scored times over the period) and stage 0.65–0.79. Total flow is over-predicted by 20 % at the scored times and 20–35 % by year, more than in the calibration years. The exceedance curves (Fig. 22) show two parts: the dry-weather and recession flows (25–90 % exceedance) are 33–47 % high (median flow 0.138 $m^3/s$ against 0.094 $m^3/s$ measured), and moderate storm flows (about 1 % of the hours) are up to 63 % high, while flows at 5 % exceedance are reproduced and the largest are within about 20 %. Stage follows the same pattern with smaller ratios (6–12 % high at low flow). Part of the excess comes from the rainfall data set (NOAA AORC), which from 2022 on is 15–29 % higher than the Reagan National Airport gauge: with the rain rescaled to the gauge-based gridMET monthly totals, the volume error of 2022–2025 falls from 20–35 % to 9–22 % and the period total from 20 to 14 %. The 25–32 % excess of 2020–2021 remains; together with the calibration years, where the rain agrees with the gauge, it shows that the model itself also keeps too much water in the stream between storms.

**Table 4:** Validation, WY2020–2025 (model run continuously from October 2013 with the calibrated parameters; values at the scored observation times).

| | NSE | logNSE | Volume ratio | Peak ratio |
|---|---|---|---|---|
| Flow | 0.75 | 0.79 | 1.20 | 1.21 |
| Stage | 0.77 | 0.79 | | |

**Table 5:** Validation by water year (hourly values): flow NSE and volume ratio with the AORC rain used by the model, and the volume ratio with the rain rescaled to gridMET.

| | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|
| Flow NSE | 0.79 | 0.72 | 0.43 | 0.54 | 0.61 | 0.87 |
| Flow logNSE | 0.64 | 0.70 | 0.72 | 0.64 | 0.76 | 0.65 |
| Volume ratio | 1.28 | 1.25 | 1.34 | 1.35 | 1.20 | 1.25 |
| Volume ratio, gridMET rain | 1.32 | 1.25 | 1.22 | 1.09 | 1.13 | 1.10 |
| Stage NSE | 0.76 | 0.79 | 0.66 | 0.65 | 0.75 | 0.70 |

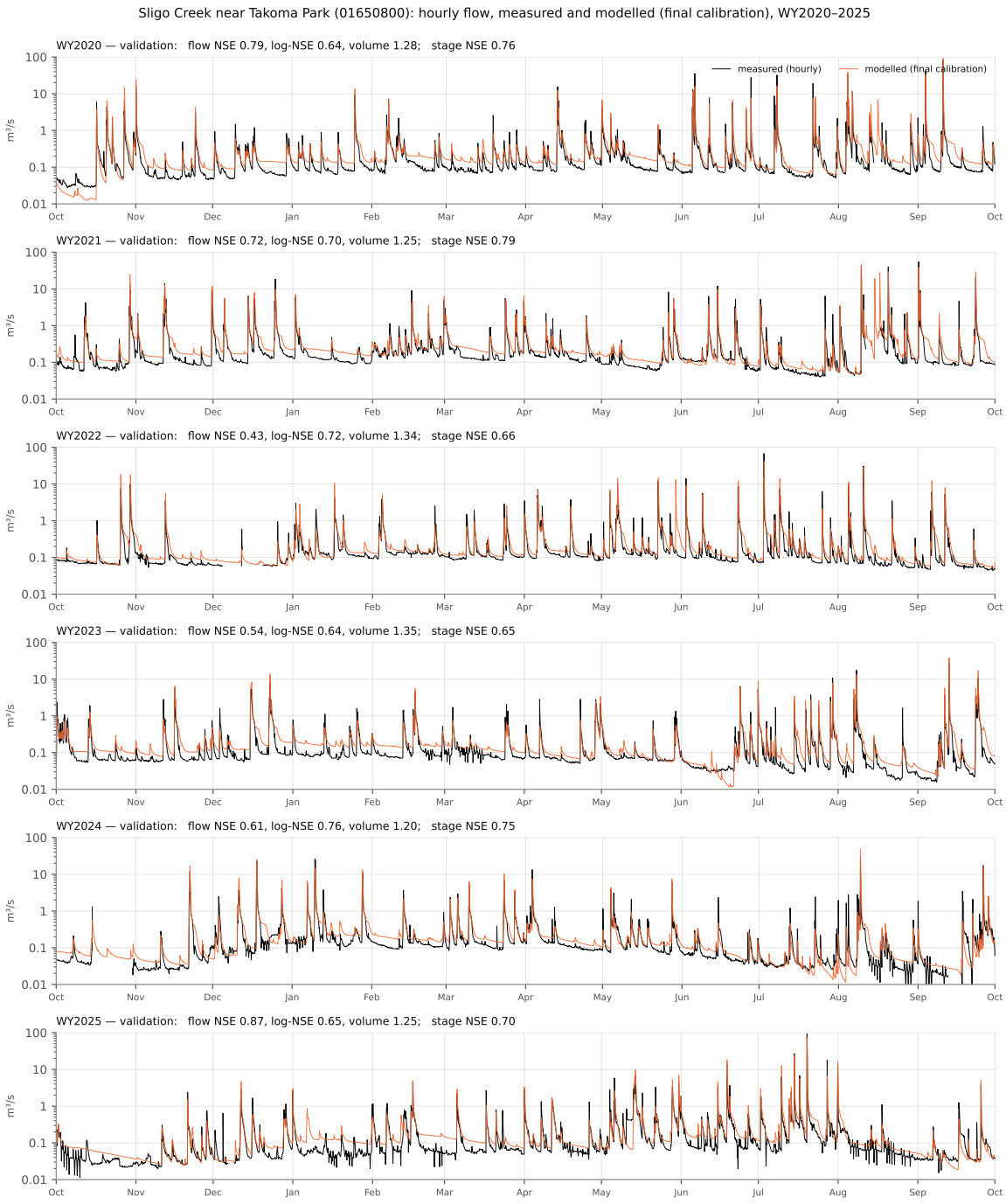



**Figure 21:** Validation: hourly flow at the USGS gage, measured and modelled, WY2020–2025.

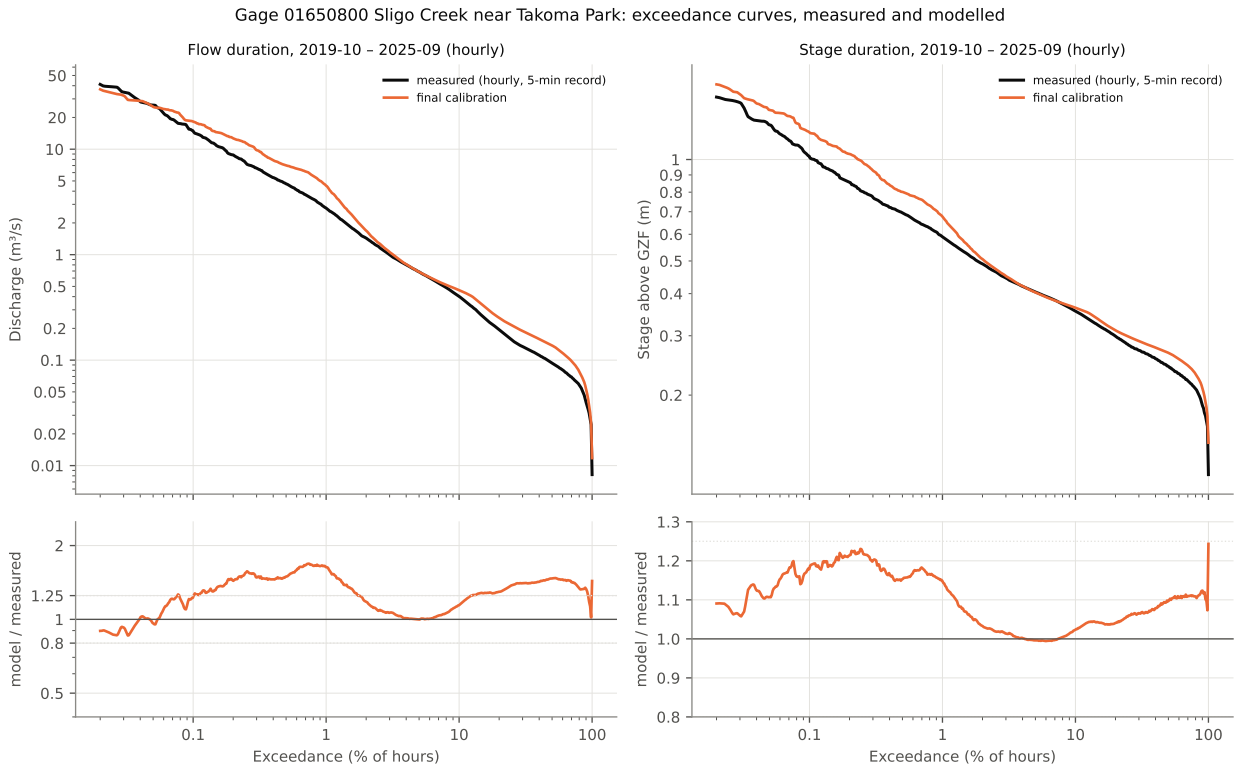



**Figure 22:** Validation: hourly flow and stage exceedance (duration) curves (log–log), WY2020–2025, and the ratio of modelled to measured values at equal exceedance.

**Fit measures.** NSE: Nash–Sutcliffe efficiency (1 is perfect, 0 is no better than the mean); logNSE: the same on the logarithms, which weights low flows and recessions; volume ratio: total model over observed flow; peak ratio: model over observed mean of the highest 20 % of flows; KGE: Kling–Gupta efficiency, which combines correlation, variability and bias (1 is perfect).

## 7.1 Comparison with the National Water Model

NOAA's National Water Model v3.0 retrospective simulation (also driven by AORC rain) was scored at the same times (Table 6 and Fig. 23). Over both periods it reproduces the gage much less well than this model, mainly because it keeps the dry-weather flow 4–5 times too high (this model: 1.2–1.7 times); the same was found for Rock Creek.

**Table 6:** This model and the National Water Model v3.0 at USGS 01650800.

| Period | Model | NSE | logNSE | KGE | Volume ratio |
|---|---|---|---|---|---|
| WY2018–2019 (calibration) | this model | 0.72 | 0.83 | 0.76 | 1.17 |
| | NWM v3.0 | 0.36 | 0.43 | 0.65 | 1.12 |
| WY2020 – Jan 2023 (validation) | this model | 0.72 | 0.80 | 0.74 | 1.21 |
| | NWM v3.0 | 0.37 | 0.23 | 0.64 | 1.17 |

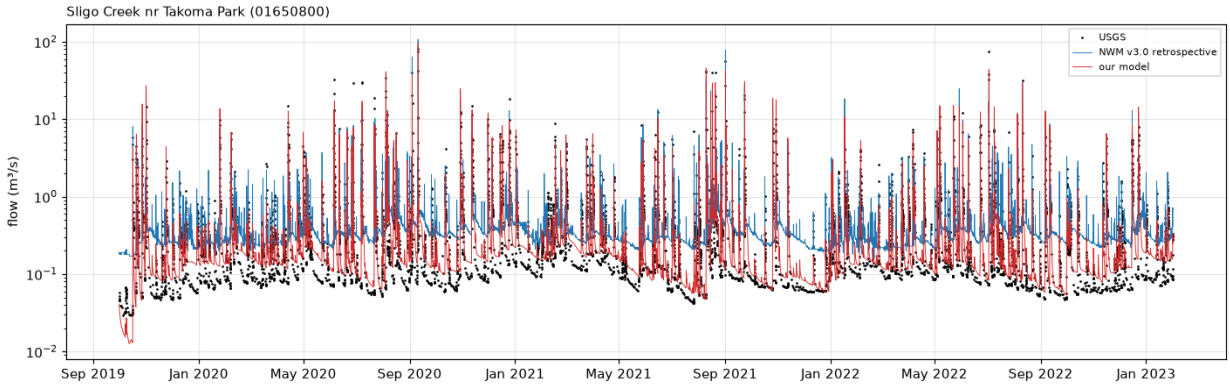



**Figure 23:** Measured flow, the National Water Model v3.0 retrospective and this model at the USGS gage, October 2019 – January 2023.

# 8 Limitations

- **One gage.** Only the upper 16.7 km$^2$ are gaged; the 12.4 km$^2$ below the gage, including Long Branch and Takoma Branch, use the watershed-wide parameters without a local check.
- **Flow volume.** The model over-predicts the total flow by 14–16 % in the calibration years and 20–35 % in the validation years, from too much flow in the recessions after storms and in moderate storms; the measured stage, which is fitted together with the flow, is reproduced much more closely, so part of the difference lies in the representation of the channel at the gage (stage for a given flow).
- **Deep-soil drainage.** The calibration makes the soil below the surface drain twenty times faster than the SSURGO values, which may stand for flow paths the model does not have, such as utility and pipe trenches and leaking pipes.
- **Uncertainty bands.** The posterior of the crop coefficients and the regional specific yield is only approximately converged. The 95 % band of the twin represents the parameters only and is much narrower than the scatter of the measurements around the model (Section 6.1); errors of the rainfall and of the model structure, and their effect on unobserved quantities such as soil moisture and groundwater, are not represented.
- **Rainfall.** The AORC rain is 15–29 % above the nearest airport gauge from 2022 on; forecasts and simulations of those years inherit this.
- **Groundwater parameters** are not uniquely determined by one gage: different combinations of regional and near-stream aquifer properties fit almost equally well.
- **Wheaton Branch pond:** the outlet dimensions are not published and were assumed.
- **Not represented:** snow, withdrawals and transfers, inflow from leaking water and sewer pipes, and changes in the watershed over time.

# 9 Software and reproducibility

The model is built and run with open-source software: OpenHydroQual (`https://github.com/ArashMassoudieh/OpenHydroQual`; Massoudieh et al., 2023; Massoudieh and Shakouri, 2025),

the digital-twin engine OpenHydroTwin (`https://github.com/ArashMassoudieh/OHTwin`) and the watershed twin and viewer OpenWatershedTwin (`https://github.com/ArashMassoudieh/OpenWatershedTwin`). Every input file of the model is produced by numbered scripts from the public data listed in Table 1. The calibrated model, with its forcing and observations, the OpenHydroQual templates it uses and a GeoPackage of every sub-catchment and channel segment with all model parameters, can be downloaded from the twin's page ("Download model"; `https://openhydrotwin.com/SligoCreek/download/SligoCreek_model.zip`, CC BY 4.0).

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